



## AQ8.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Which of these sets form an orthonormal basis of  $R^3$  with respect to the dot product as the inner product in  $R^3$ ?

☐

$\{(1, 0, 0), (0, 1, 0)\}$

☐

$\{(2, 0, 0), (0, 2, 0), (0, 0, 2)\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(2, 1, -1), (-1, -1, 1), (1, 2, -2)\}$

☐

$\{(2, 1, -1), (-1, -1, 1), (1, 2, -2)\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(1, \sqrt{6}, 1), (\sqrt{6}, 1, 1), (1, 1, \sqrt{6})\}$

☐

$\{(2, 1, -1), (-1, -1, 1), (1, 2, -2)\}$

☐

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

☐

$\{(1, 0, 1), (0, 1, 0), (1, 0, 1)\}$

☐

$\{(1, 0, 1), (0, 1, 0), (1, 0, 1)\}$

☐

$\{(1, 0, 1), (0, 1, 0), (1, 0, 1)\}$

☐

$\{(1, 0, 1), (0, 1, 0), (1, 0, 1)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$



$\sqrt{1, 6}$ 
 $\sqrt{1, 6}$ 
 $\sqrt{2), (11}$ 
 $\sqrt{-3, 11}$ 
 $\sqrt{1, 11}$ 
 $\sqrt{1), (66}$ 
 $\sqrt{1, 66}$ 
 $\sqrt{7, 66}$ 
 $\sqrt{-4)\}$ 
 $\{(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\} \{(\sqrt{2}$ 
 $\sqrt{1, 0, \sqrt{2}}$ 
 $\sqrt{-1), (0, 1, 0), (\sqrt{2}$ 
 $\sqrt{1, 0, \sqrt{2}}$ 
 $\sqrt{1)\}$ 

1 point

Consider two orthogonal vectors  $a, b$  in  $\mathbb{R}^2$ . If  $a + b$  and  $a - b$  are orthogonal, then choose the correct option.

☐

$\|a\| = \|b\| = 1$

☐

$\|a\| = \|b\|$

☐

$\|a\| = 2\|b\|$

☐

$2\|a\| = \|b\|$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\|a\| = \|b\|$

1 point

Consider  $a = (1, 1)$ ,  $b = (1, -1)$ . Let  $V = \text{Span}\{a, b\}$ . Choose the correct options by considering the standard inner product (dot product).

☐

Vectors  $a, b$  form an orthogonal basis for  $V$ .

☐

Vectors  $a, b$  form an orthonormal basis for  $V$ .

☐

There exist scalar multiples of  $a, b, c$  which form an orthonormal basis

☐

There do not exist scalar multiples of  $a, b, c$  which form an orthonormal basis

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Vectors  $a, b, c$  form an orthogonal basis for  $V$ .

There exist scalar multiples of  $a, b, c$  which form an orthonormal basis

## Level 2

1 point

Choose the set of correct statements.

☐

The determinant of a matrix formed by 3 orthonormal vectors in  $\mathbb{R}^3$  is  $\pm 1$ .

☐

The determinant of a matrix formed by 3 orthonormal vectors in  $\mathbb{R}^3$  is 0.

☐

The determinant of a matrix formed by 2 orthonormal vectors in  $\mathbb{R}^2$  is  $\pm 1$ .

☐

The determinant of a matrix formed by 2 orthonormal vectors in  $\mathbb{R}^2$  is 0.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The determinant of a matrix formed by 3 orthonormal vectors in  $\mathbb{R}^3$  is  $\pm 1$ .

The determinant of a matrix formed by 2 orthonormal vectors in  $\mathbb{R}^2$  is  $\pm 1$ .

1 point

Consider a system of linear equations:

$$2x_1 + 2x_2 + 7x_3 = b_1 \quad 2x_1 + 2x_2 + 7x_3 = b_1$$

$$2x_1 + x_2 - 10x_3 = b_2 \quad 2x_1 + x_2 - 10x_3 = b_2$$

$$3x_1 - 2x_2 + 2x_3 = b_3 \quad 3x_1 - 2x_2 + 2x_3 = b_3.$$

Let  $A$  be the coefficient matrix of the given system of linear equations. Let a matrix  $B$  contain the column vectors of  $A$ , which are normalized by their respective norms, as its columns (i.e. first column vector of  $A$  normalized by its norm is the first column of  $B$ ). Which of the following statements are true ?

☐

The determinant of  $BB^T$  is 11.

☐

$BB^T$  is an identity matrix.

☐

$BB^T$  is a scalar matrix.

☐

$BB^T$  is a diagonal matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The determinant of  $BB^T$  is 11.

$BB^T$  is an identity matrix.

$BB^T$  BBT is a scalar matrix.

$BB^T$  BBT is a diagonal matrix.

1 point

Choose the correct option(s).

- ☐ The vectors in an orthonormal set are linearly independent.
- ☐ A set of linearly dependent vectors can be orthonormal.
- ☐ A set of linearly independent vectors is always orthonormal.
- ☐ A set of linearly independent vectors is always orthogonal but not orthonormal.
- ☐ The vectors in an orthogonal set are linearly independent.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The vectors in an orthonormal set are linearly independent.

The vectors in an orthogonal set are linearly independent.

Consider the inner product  $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$   $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$  where

$u = (u_1, u_2)$ ,  $v = (v_1, v_2)$   $u = (u_1, u_2)$ ,  $v = (v_1, v_2)$  are vectors in  $R^2$ . Answer questions 7 and 8 based on this information.

1 point

Which of the following is an orthonormal basis with respect to the inner product defined above?

☐

$\{(2, 3), (-4, 4)\}$

☐

$\{\frac{1}{\sqrt{30}}(2, 3), \frac{1}{\sqrt{80}}(-4, 4)\}$

☐

$\{(2, 3), (-4, 4)\}$

☐

$\{(-4, 4)\}$

☐

$\{\frac{1}{\sqrt{13}}(2, 3), \frac{1}{\sqrt{32}}(-4, 4)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{(-4, 4)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{\frac{1}{\sqrt{13}}(2, 3), \frac{1}{\sqrt{13}}(-3, 2)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{(-3, 2)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{\frac{1}{\sqrt{30}}(2, 3), \frac{1}{\sqrt{80}}(-4, 4)\}$

☐

$\{(2, 3), (-4, 4)\}$

☐

$$1(-4, 4)\}$$

1 point

Use the orthonormal basis  $\{u, v\}$  obtained in question 7 with respect to the defined inner product.

Express the vector  $(4, 0)$  as a linear combination of the basis vectors  $u$  and  $v$ , as

$(4, 0) = c_1 u + c_2 v$ . Which of the following gives the coefficients of the linear combination?

☐

$$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}} c_1 = 30$$

☐

$$24, c_2 = 80$$

☐

$$-48$$

☐

$$c_1 = \frac{24}{\sqrt{13}}, c_2 = \frac{-48}{\sqrt{32}} c_1 = 13$$

☐

$$24, c_2 = 32$$

☐

$$-48$$

☐

$$c_1 = \frac{8}{\sqrt{13}}, c_2 = \frac{-16}{\sqrt{32}} c_1 = 13$$

☐

$$8, c_2 = 32$$

☐

$$-16$$

☐

$$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{48}{\sqrt{80}} c_1 = 30$$

☐

$$24, c_2 = 80$$

☐

$$48$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}} c_1 = 30$$

☐

$$24, c_2 = 80$$

☐

$$-48$$

[Check Answers](#)

Your score is: 0/8